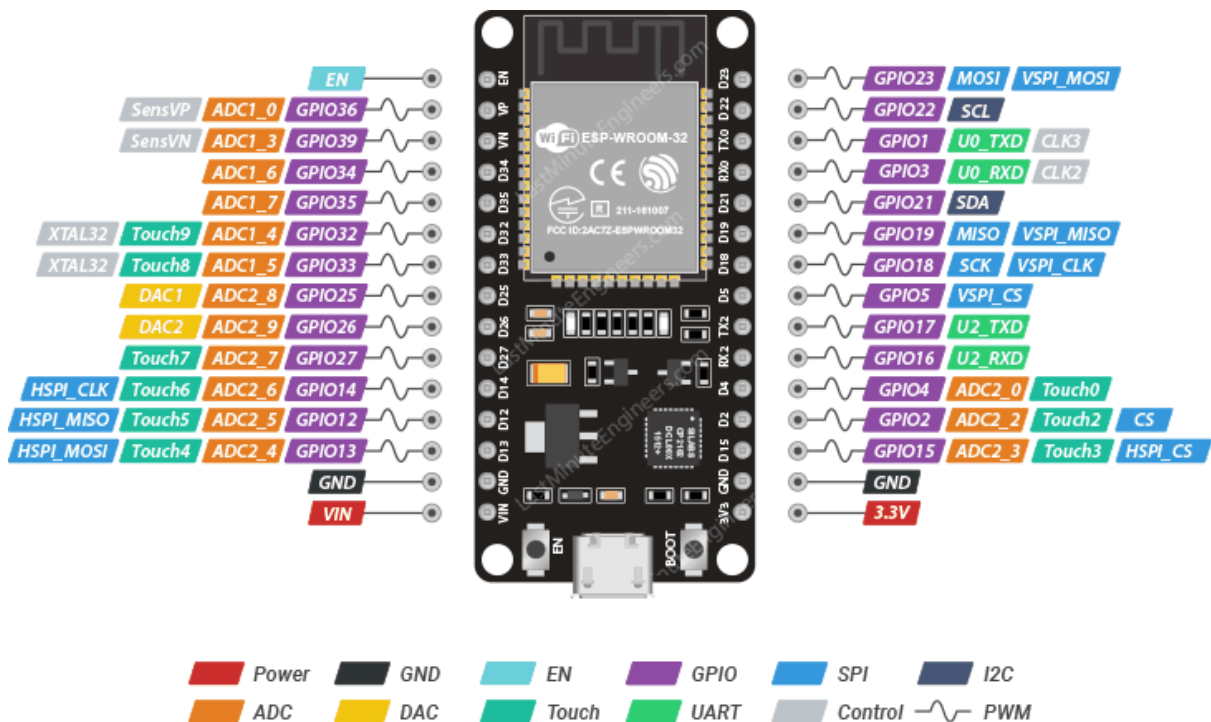


AI Based Smart System Development (Real Time Hardware Implementation)

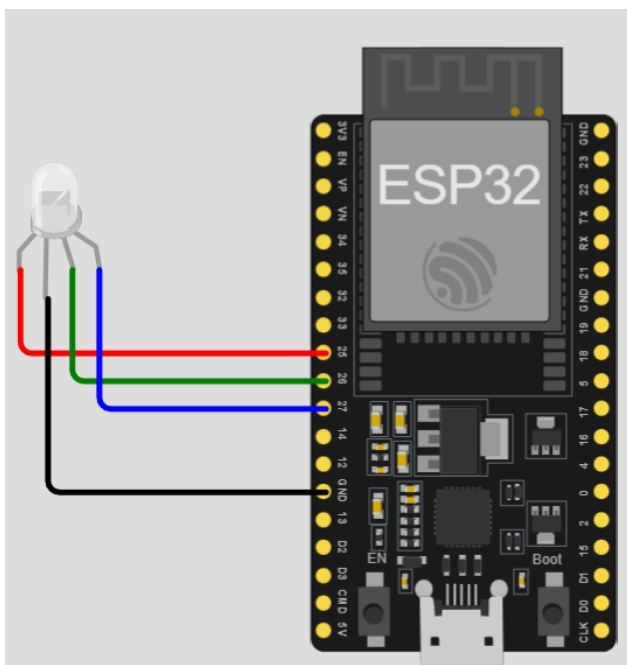
Week – 1 Lecture

<https://lastminuteengineers.com/esp32-pinout-reference/>



- 1) Create a setColor() function that accepts Red, Green, and Blue brightness values (0–255) and then use analogWrite() to generate PWM signals.

RGB LED Connections



RGB LED Pin	ESP32 Pin
Red	GPIO 25
Green	GPIO 26
Blue	GPIO 27
Common Cathode	GND

AI Based Smart System Development (Real Time Hardware Implementation)

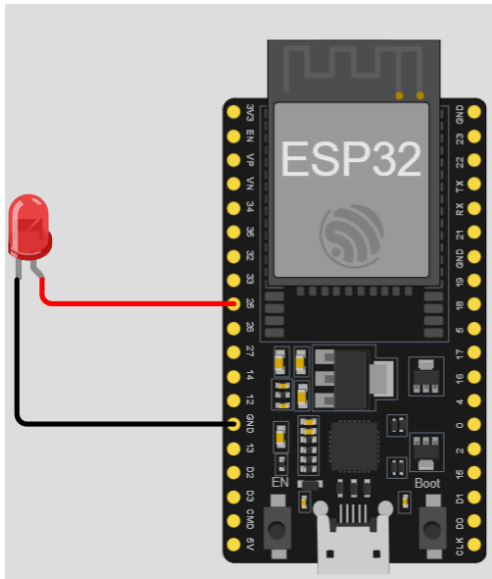
Week – 1 Lecture

```
1 // RGB LED Pins
2 const int RED_PIN = 25;
3 const int GREEN_PIN = 26;
4 const int BLUE_PIN = 27;
5
6 // Function to set RGB color
7 void setColor(int red, int green, int blue)
8 {
9     analogWrite(RED_PIN, red);
10    analogWrite(GREEN_PIN, green);
11    analogWrite(BLUE_PIN, blue);
12 }
13
14 void setup()
15 {
16    pinMode(RED_PIN, OUTPUT);
17    pinMode(GREEN_PIN, OUTPUT);
18    pinMode(BLUE_PIN, OUTPUT);
19 }
20
21 void loop()
22 {
23     // Red
24     setColor(255, 0, 0);
25     delay(1000);
26
27     // Green
28     setColor(0, 255, 0);
29     delay(1000);
30
31     // Blue
32     setColor(0, 0, 255);
33     delay(1000);
34
35     // Yellow
36     setColor(255, 255, 0);
37     delay(1000);
38
39     // Cyan
40     setColor(0, 255, 255);
41     delay(1000);
42
43     // Magenta
44     setColor(255, 0, 255);
45     delay(1000);
46
47     // White
48     setColor(255, 255, 255);
49     delay(1000);
50
51     // Orange
52     setColor(255, 100, 0);
53     delay(1000);
54
55     // Purple
56     setColor(128, 0, 255);
57     delay(1000);
58
59     // Pink
60     setColor(255, 50, 100);
61     delay(1000);
62
63     // Off
64     setColor(0, 0, 0);
65     delay(1000);
66 }
67
```

AI Based Smart System Development (Real Time Hardware Implementation)

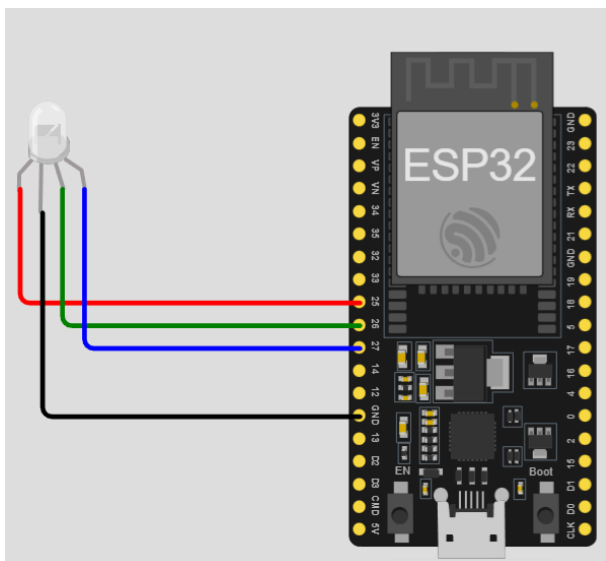
Week – 1 Lecture

2) PWM controls brightness.



```
1  const int RED_PIN = 25;
2
3  void setup()
4  {
5      pinMode(RED_PIN, OUTPUT);
6  }
7
8  void loop()
9  {
10     // Increase brightness
11     for(int brightness = 0; brightness <= 255; brightness++)
12     {
13         analogWrite(RED_PIN, brightness);
14         delay(10);
15     }
16
17     // Decrease brightness
18     for(int brightness = 255; brightness >= 0; brightness--)
19     {
20         analogWrite(RED_PIN, brightness);
21         delay(10);
22     }
23 }
```

3) RGB Brightness Control

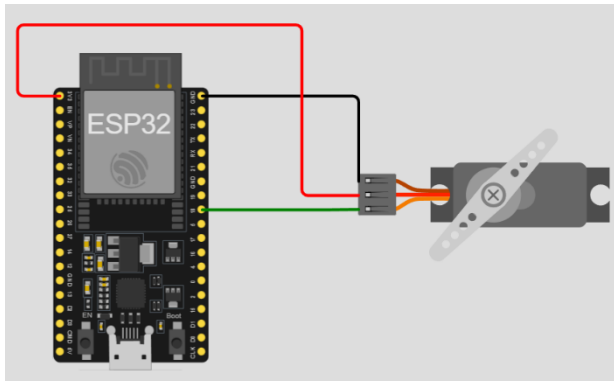


```
1  const int RED_PIN  = 25;
2  const int GREEN_PIN = 26;
3  const int BLUE_PIN  = 27;
4
5  void setColor(int red, int green, int blue)
6  {
7      analogWrite(RED_PIN, red);
8      analogWrite(GREEN_PIN, green);
9      analogWrite(BLUE_PIN, blue);
10 }
11
12 void setup()
13 {
14     pinMode(RED_PIN, OUTPUT);
15     pinMode(GREEN_PIN, OUTPUT);
16     pinMode(BLUE_PIN, OUTPUT);
17 }
18
19 void loop()
20 {
21     // Red brightness variation
22     for(int i = 0; i <= 255; i += 25)
23     {
24         setColor(i, 0, 0);
25         delay(500);
26     }
27
28     // Green brightness variation
29     for(int i = 0; i <= 255; i += 25)
30     {
31         setColor(0, i, 0);
32         delay(500);
33     }
34
35     // Blue brightness variation
36     for(int i = 0; i <= 255; i += 25)
37     {
38         setColor(0, 0, i);
39         delay(500);
40     }
41 }
```

AI Based Smart System Development (Real Time Hardware Implementation)

Week – 1 Lecture

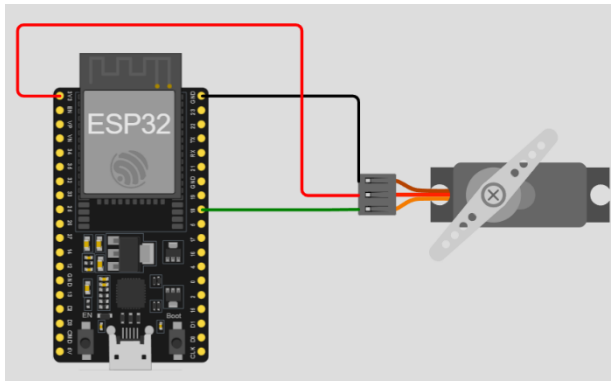
4) Move your Servo with Different angle



```
1 #include <ESP32Servo.h>
2
3 Servo myServo;
4
5 void setup()
6 {
7   myServo.attach(18);
8 }
9
10 void loop()
11 {
12   myServo.write(0);
13   delay(1000);
14
15   myServo.write(45);
16   delay(1000);
17
18   myServo.write(90);
19   delay(1000);
20
21   myServo.write(135);
22   delay(1000);
23
24   myServo.write(180);
25   delay(1000);
26 }
```

```
1 #include <ESP32Servo.h>
2
3 Servo myServo;
4
5 void setAngle(int angle)
6 {
7   myServo.write(angle);
8 }
9
10 void setup()
11 {
12   myServo.attach(18);
13 }
14
15 void loop()
16 {
17   setAngle(0);
18   delay(1000);
19
20   setAngle(45);
21   delay(1000);
22
23   setAngle(90);
24   delay(1000);
25
26   setAngle(135);
27   delay(1000);
28
29   setAngle(180);
30   delay(1000);
31 }
```

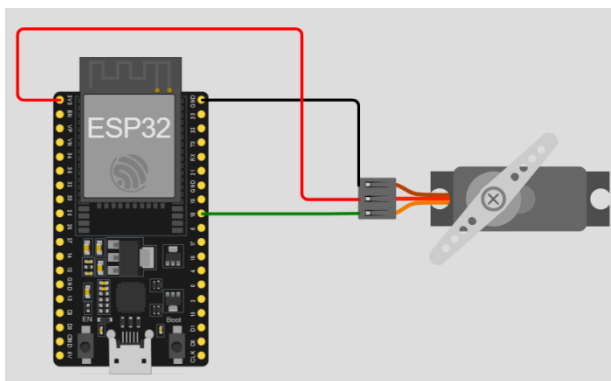
5) Smooth Servo Movement (Instead of jumping directly)



```
1 #include <ESP32Servo.h>
2
3 Servo myServo;
4
5 void setup()
6 {
7   myServo.attach(18);
8 }
9
10 void loop()
11 {
12   for(int angle=0; angle<=180; angle++)
13   {
14     myServo.write(angle);
15     delay(15);
16   }
17
18   for(int angle=180; angle>=0; angle--)
19   {
20     myServo.write(angle);
21     delay(15);
22   }
23 }
```

Increase delay(15) to delay(50) for slow Motion

6) Create a Reusable Function

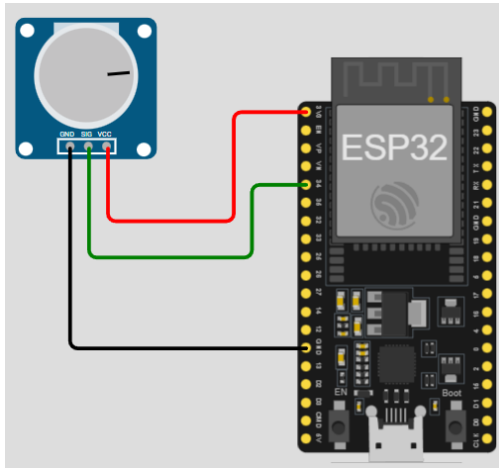


```
1 #include <ESP32Servo.h>
2
3 Servo myServo;
4
5 void moveServo(int startAngle,
6               int endAngle,
7               int speedDelay)
8 {
9   if(startAngle < endAngle)
10   {
11     for(int angle=startAngle;
12         angle<=endAngle;
13         angle++)
14     {
15       myServo.write(angle);
16       delay(speedDelay);
17     }
18   }
19   else
20   {
21     for(int angle=startAngle;
22         angle>=endAngle;
23         angle--)
24     {
25       myServo.write(angle);
26       delay(speedDelay);
27     }
28   }
29 }
30
31 void setup()
32 {
33   myServo.attach(18);
34 }
35
36 void loop()
37 {
38   moveServo(0,180,10);
39
40   delay(1000);
41
42   moveServo(180,0,30);
43
44   delay(1000);
45 }
```

AI Based Smart System Development (Real Time Hardware Implementation)

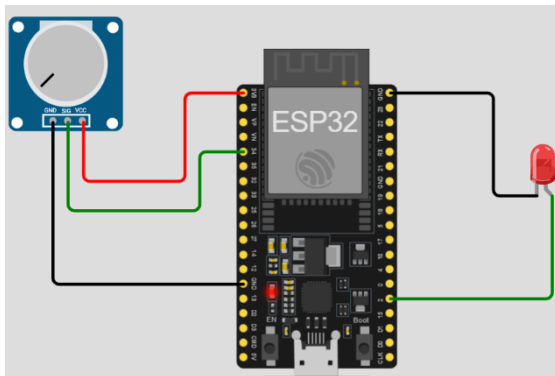
Week – 1 Lecture

7) Reading a Potentiometer with ESP32



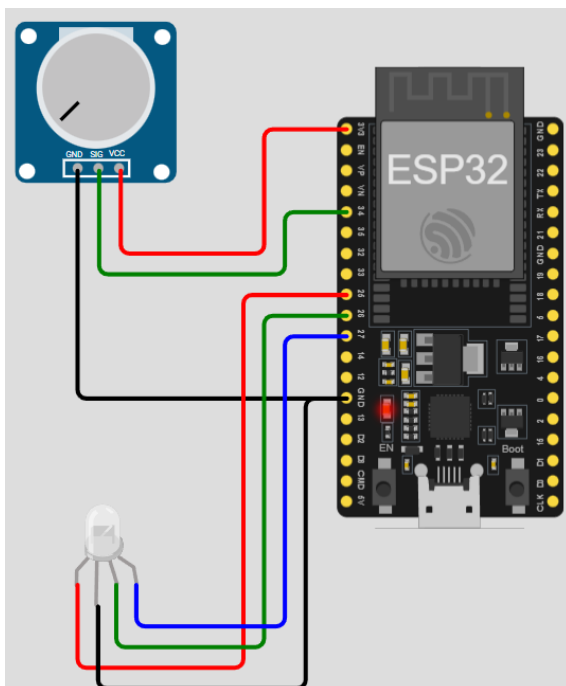
```
1  const int potPin = 34;
2
3  void setup()
4  {
5      Serial.begin(115200);
6  }
7
8  void loop()
9  {
10     int potValue = analogRead(potPin);
11
12     Serial.print("Potentiometer Value = ");
13     Serial.println(potValue);
14
15     delay(1000);
16 }
```

8) Control LED Brightness using Pot



```
1  const int potPin = 34;
2  const int ledPin = 2;
3
4  void setup()
5  {
6      pinMode(ledPin, OUTPUT);
7  }
8
9  void loop()
10 {
11     int potValue = analogRead(potPin);
12
13     int brightness = map(potValue,
14                         0,4095,
15                         0,255);
16
17     analogWrite(ledPin, brightness);
18
19     delay(10);
20 }
```

9) Control RGB LED Color using Pot

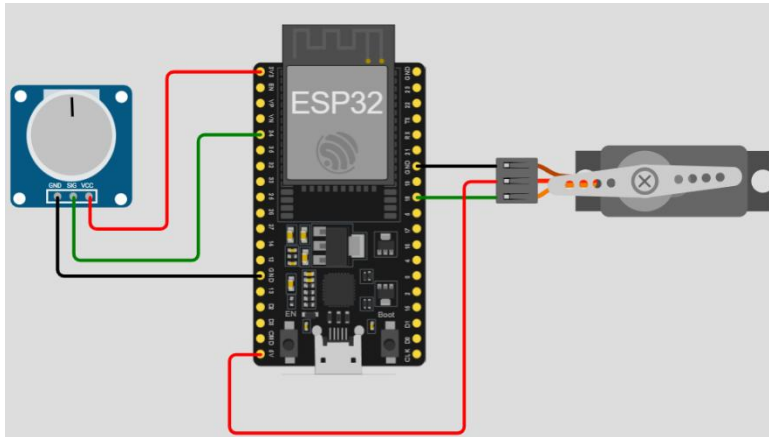


```
1  const int potPin = 34;
2
3  void setColor(int r,int g,int b)
4  {
5      analogWrite(25,r);
6      analogWrite(26,g);
7      analogWrite(27,b);
8  }
9
10 void setup()
11 {
12 }
13
14 void loop()
15 {
16     int potValue = analogRead(34);
17
18     int brightness =
19         map(potValue,0,4095,0,255);
20
21     setColor(brightness,0,0);
22     //setColor(0,brightness,0);
23     //setColor(0,0, brightness);
24
25     delay(10);
26 }
```

AI Based Smart System Development (Real Time Hardware Implementation)

Week – 1 Lecture

10) Control Servo with Potentiometer



```
1  #include <ESP32Servo.h>
2
3  Servo myServo;
4
5  const int potPin = 34;
6
7  void setup()
8  {
9      myServo.attach(18);
10
11     Serial.begin(115200);
12 }
13
14 void loop()
15 {
16     int potValue = analogRead(potPin);
17
18     int angle =
19         map(potValue,
20             0,4095,
21             0,180);
22
23     myServo.write(angle);
24
25     Serial.print("Angle = ");
26     Serial.println(angle);
27
28     delay(20);
29 }
```